

AlgoLend Platform: Technical & Legal Breakdown

This document provides a high-level technical breakdown of the AlgoLend system's core modules, data flows, and third-party integrations. It is structured to help identify potential compliance gaps (e.g., under the National Credit Act (NCA), Protection of Personal Information Act (POPIA), and Electronic Communications and Transactions Act (ECTA) in South Africa).

1. Client Onboarding & Identity Verification (KYC/AML)

Overview: The system collects Personal Identifiable Information (PII) to verify user identity, assess risk, and prevent fraud.

Key Technical Actions:

- **Data Collection:** Full name, ID number, contact details, physical/work address, employment details.
- **Biometric & Liveness Checks:** The system integrates with **TruID** to perform real-time biometric liveness checks and facial comparison against official databases (e.g., Home Affairs).
- **Company Checks:** Real-time CIPC company checks and director screening for corporate entities or SME lending.

[!WARNING]

Legal Considerations:

- **POPIA:** Is explicit consent obtained before sharing ID numbers/selfies with TruID and Home Affairs? Is data retention minimized after verification?
- **AML/FICA:** Does the biometric and CIPC screening satisfy all Financial Intelligence Centre Act (FICA) requirements for establishing business relationships?

2. Financial Profiling & Credit Scoring

Overview: The system evaluates the applicant's affordability and creditworthiness to make loan decisions.

Key Technical Actions:

- **Bank Statement Scraping/Open Banking:** Uses **TruID / Banking APIs** to programmatically access and analyze the user's transaction history and income.
- **Credit Bureau Integrations:** The platform queries credit bureaus to pull credit scores, existing debt obligations, and default history.
- **Automated Risk Engine:** A backend rules engine processes the financial data to output a risk score, risk-based pricing, and automated approval/rejection recommendations.

[!IMPORTANT]

Legal Considerations:

- **NCA Affordability Assessments:** Does the algorithmic evaluation strictly adhere to the NCA's required affordability assessment regulations?
- **Consent:** Are users explicitly consenting to bank account access (screen scraping/Open Banking) and credit bureau inquiries?
- **Automated Decision Making:** Under POPIA, data subjects have rights regarding automated processing that produces legal effects. Are there mechanisms for manual review or appeal if a loan is algorithmically denied?

3. Contracting & E-Signatures

Overview: Generation and signing of the legally binding credit agreement.

Key Technical Actions:

- **Dynamic Document Generation:** Merges user data, loan terms (principal, interest, credit life, fees), and NCR disclosures into a PDF contract.
- **E-Signatures:** Integrates with **DocuSeal** to capture the borrower's electronic signature and issue the finalized "AlgoLend Small Credit Agreement."

[!WARNING]

Legal Considerations:

- **ECTA Compliance:** Does the DocuSeal implementation meet the requirements for an Advanced Electronic Signature (AES) if required, or is a standard electronic signature legally sufficient for these specific credit agreements?
- **NCA Disclosures:** Ensure all mandatory pre-agreement statements and quotations are presented to the user *before* the contract is signed, with adequate cooling-off periods if applicable.

4. Payment Collection & Mandate Control

Overview: How AlgoLend collects loan repayments from the borrower's bank account.

Key Technical Actions:

- **DebiCheck Mandates:** Integrates with **SureSystems** to register and load DebiCheck mandates (e.g., authenticated early debit orders).
- **Mandate Workflows:** The system sends authorization requests directly to the user's banking app/USSD for approval before collections can commence.

[!IMPORTANT]

Legal Considerations:

- **PASA & DebiCheck Rules:** Is the mandate creation process strictly compliant with Payments Association of South Africa (PASA) rules? What is the legal fallback if a user fails to authenticate the mandate on their phone?

5. Data Hosting & Cloud Infrastructure

Overview: The physical and logical storage of user and financial data.

Key Technical Actions:

- **Supabase (PostgreSQL):** Acts as the primary database, handling authentication, role-based access control (RBAC), and storage of all application data.
- **Communications:** Integrates with **Resend** (email) and **Web-Push** (browser notifications) to send transactional updates and contract copies.

[!CAUTION]

Legal Considerations:

- **Cross-Border Data Transfers:** Supabase is typically hosted on AWS/GCP. If the data is hosted outside of South Africa, POPIA requires the destination country to have adequate data protection laws, or the user must explicitly consent to cross-border transfer.
- **Data Security:** Ensure encryption in transit and at rest is legally sufficient to protect highly sensitive financial profiles and banking credentials.
- **Marketing Consent:** If Resend is used for marketing (not just transactional emails), opt-in/opt-out mechanisms must be strictly enforced.

Summary of Third-Party Dependencies

Any legal review of AlgoLend's terms and conditions should explicitly cover data sharing with these specific subprocessors:

1. **Supabase:** Core database and authentication.
2. **TruID:** KYC, Biometrics, and Banking data access.
3. **Credit Bureaus:** (Specific bureau name used by the API).
4. **DocuSeal:** E-signature and contract generation.
5. **SureSystems:** DebiCheck mandate processing.
6. **Resend:** Email communications.